FATEQ TECHNOLOGIES | INDEPENDENT PLM CONSULTING

CAD-to-ERP Digital Thread Integration & Automated EBOM Synchronization

A Technical Case Study on Siemens Teamcenter 2606, Active Integration Gateway (AIG), and SAP ERP Integration

1. Executive Summary

This technical case study presents the end-to-end architecture, configuration, and validation playbook for integrating 3D Mechanical CAD (Siemens NX) with Enterprise Resource Planning (SAP ERP) via Siemens Teamcenter 2606 and Active Integration Gateway (AIG / T4S). Designed by FATEQ Technologies, this implementation blueprint solves persistent shop-floor data corruption, manual CSV export bottlenecks, duplicate material master creation, and unaligned Engineering Bills of Materials (EBOM) versus Manufacturing BOMs (MBOM). By establishing an automated, bi-directional digital thread, the solution enforces pre-transfer data validation, ECLASS CST classification checks, asynchronous transaction queueing, and automated error recovery workflows.

Key Technical Outcome: Achieved 99.8% transactional success rate across 15,000+ monthly material master and BOM synchronization transactions, eliminated manual data entry efforts by 85%, and enforced zero duplicate material creation across multi-site ERP plants.

2. Business Problem & As-Is Operational Baseline

In the baseline operational state, engineering teams authored CAD models in Siemens NX and manually exported BOM structures into flat CSV files. Manufacturing planners then manually re-keyed these attributes into SAP ERP to create Material Masters (MARA/MAKT) and Plant BOMs (STKO/STPO). This disconnected handover introduced four critical enterprise vulnerabilities:

Operational Area	Legacy Friction & Root Cause	Business & Cost Impact
Data Quality	CAD properties (material, mass, supplier P/N) manually re-typed into ERP.	35% data discrepancy rate between CAD drawing and ERP material master.
Duplicate Parts	Designers created new part numbers without checking existing ERP inventory.	Estimated $4,500 qualification and inventory holding cost per duplicate part.
Change Execution	Engineering Change Notices (ECN) approved in PLM but uncommunicated to ERP.	Shop floor assembled 200+ units using obsolete revision parts (scrapped).
Transaction Risk	Lack of transactional rollback; partial BOM transfers locked SAP queues.	Average of 18 technical support hours wasted weekly resolving batch queue hangs.

3. Project Scope & Technical Objectives

The project scope defined a fully automated, event-driven integration framework between Siemens Teamcenter 2606 Active Workspace and SAP ERP using the Siemens Active Integration Gateway (AIG) / Teamcenter Integration Framework (TcIF).

In-Scope Architecture:
• Bi-directional attribute synchronization between NX CAD properties, Teamcenter Item Revisions, and SAP Material Masters.
• Automated EBOM structure validation (mass, status, material classification) prior to triggering ERP release workflows.
• AIG / T4S mapping procedure configuration (T4S_MAPPING_MATERIAL_MASTER, T4S_MAPPING_BOM_HEADER).
• Automated error queue handling with active notification workflows to PLM data stewards.
• Scheduled night-way delta reconciliation scripts between Teamcenter database read-replicas and SAP OData endpoints.

4. Technical Challenges & Root Cause Analysis

To establish a robust integration architecture, FATEQ Technologies conducted a root cause analysis on common CAD-to-ERP failure modes:

Technical Failure Mode	Root Cause Analysis	Architectural Resolution
Attribute Schema Mismatch	CAD string properties (PART_MATERIAL) exceeded SAP field length limits.	Configured BMIDE property transformers and LOV validation pick-lists.
Zero-Mass Part Transfer	CAD models lacked assigned material densities, exporting 0.0 kg mass.	Enforced 5-level mass precedence rules and workflow release validation gates.
Queue Deadlocks	Simultaneous release of multi-level BOMs caused SAP locks on parent items.	Implemented asynchronous AIG batch queues with dependency-ordered child-first transfer.
Duplicate Material Creation	Lack of pre-transfer classification lookup in Teamcenter.	Integrated ECLASS CST classification IRDI search mandatory check in workflow.

5. Detailed Solution Design & Mapping Framework

The solution design centers on five core technical pillars that guarantee data integrity and platform stability:

A. CAD-to-Teamcenter Attribute Mapping (cad_to_tc_attribute_map.xsd):
NX CAD properties are synchronized bi-directionally with Teamcenter BMIDE Item Revision attributes upon save. The mapping schema enforces strict data type conversions and character limits:

NX CAD Property	Teamcenter BMIDE Attribute	SAP ERP Field	Validation / Transformation Rule
DB_PART_NO	item_id (Item)	MATNR	String(18), Auto-generated sequence, Non-editable
DB_PART_REV	item_revision_id	REVLV	String(2), Uppercase alphanumeric (A, B, C...)
PART_MATERIAL	nis4MaterialSpec	FERTH	LOV validation against IMM approved catalog
CALC_WEIGHT	nis4CalculatedMass	BRGEW	Double (kg), 5-level mass precedence enforcement
SUPPLIER_PN	vendor_part_number	MFRPN	Checked against Supplier Quality Hub registry

B. Teamcenter EBOM Validation & Workflow Release Gates:

Prior to initiating the AIG transfer, the Teamcenter Workflow Designer executes an automated pre-flight rule handler (EPM-check-action-handler). The workflow verifies that: (1) All BOM line items hold TCM Released status, (2) No parts contain zero mass, (3) All standard hardware items are classified in ECLASS CST, and (4) Secondary 3D JT and PDF viewable datasets are fully attached.

C. AIG / T4S Data Transfer & Exception Queueing:
The Active Integration Gateway parses the validated Teamcenter BOM structure into structured TCL/XML payloads. The mapping procedures (T4S_MAPPING_MATERIAL_MASTER and T4S_MAPPING_BOM_HEADER) execute remote function calls (RFC) to SAP. If a network or database lock occurs, the transaction is logged in the AIG Job Monitor and queued for automated retry (up to 3 attempts) before triggering a task escalation to the PLM System Administrator.

6. Implementation Phases & Teamcenter Components Used

Phase & Lifecycle Step	Core Technical Activities Performed	Teamcenter Components Utilized
Phase 1: Data Model Setup	Deployed BMIDE custom business objects, LOVs, and naming rules.	BMIDE, Active Workspace XRT
Phase 2: Integration Mapping	Configured AIG/T4S mapping scripts, TCL transformations, and RFC endpoints.	Active Integration Gateway, T4S
Phase 3: Workflow Automation	Built ECN release workflow templates with pre-transfer validation handlers.	Workflow Designer, Dispatcher
Phase 4: Reconciliation Setup	Deployed scheduled nightly reconciliation scripts and Power BI dashboards.	OData v4 REST API, Read Replica
Phase 5: Cutover & Go-Live	Executed 7-step Deployment Center deployment with HashiCorp Vault encryption.	Deployment Center, Vault

7. Key Performance Indicators & Business Outcomes

The technical implementation delivered dramatic operational improvements across engineering and manufacturing teams. Note: Metrics are derived from illustrative reference benchmarks.

Performance Indicator (KPI)	Baseline (As-Is)	Target State	Achieved Result	Business Outcome
BOM Transfer Turnaround	45 mins (Manual)	< 2 mins (Auto)	1.2 mins	85% reduction in administrative lead time.
Data Accuracy Rate	65.0%	> 99.0%	99.8%	Near-zero shop-floor assembly rework.
Duplicate Part Creation	12% of new parts	< 2.0%	0.4%	$380K annual inventory cost avoidance.
ECO-to-ERP Release Cycle	18 Days	< 3 Days	1.5 Days	91% faster engineering change execution.

8. Lessons Learned & Strategic Recommendations

• Enforce Data Quality at the Source: Validating CAD mass and material properties inside NX prior to check-in eliminates 90% of downstream ERP transfer errors.
• Prefer OOTB Configuration Over Custom Code: Utilizing standard AIG / T4S mapping procedures instead of custom C++/ITK integrations reduces platform upgrade costs by over 60%.
• Implement Dependency-Ordered Queueing: Transferring child components before parent assembly BOMs prevents SAP ERP structural deadlock errors.

9. Quality Control & Mandatory Disclaimer

Quality Control Gate	Status	Verification Details
Technical Architecture Review	PASSED	Verified against Siemens Teamcenter 2606 & AIG technical docs.
Data Model & Schema Audit	PASSED	Validated BMIDE properties and cad_to_tc_attribute_map schema.
Security & Compliance Gate	PASSED	Credentials encrypted via HashiCorp Vault; no flat file passwords.

MANDATORY DISCLAIMER:

This report is an illustrative technical portfolio sample produced by FATEQ Technologies (Prepared by Syed Abdul Hairu). Illustrative scenarios, architecture models, metrics, and outcomes are hypothetical and do not represent a completed customer engagement. Actual client implementations require site-specific evidence, official software release verification, and tailored solution validation.